

PRIYANKA KUMARI

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer building production AI systems - agentic pipelines, RAG and LLM applications - together with the backend engineering that keeps them correct under load. Currently AI Engineer at ShipCube, promoted from a data science internship after three months, delivering internal AI and automation services for 3PL logistics operations. Experienced across Generative AI, NLP, computer vision and high-concurrency API design in Python, with independent projects spanning agentic research, generative video and concurrency-critical backends.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript

Generative AI and NLP: LLMs, Agentic AI, RAG, LangChain, LangGraph, Hugging Face Transformers, ChromaDB, BM25, Vertex AI, Gemini, Prompt Engineering

Machine Learning: scikit-learn, TF-IDF, Classification, Recommendation Systems, Feature Engineering, NLTK, Sentiment Analysis

Computer Vision: OpenCV, MediaPipe, DeepFace, Real-time inference

Backend and Data: FastAPI, Flask, Streamlit, Node.js, Express, PostgreSQL, MongoDB, Redis, BigQuery, Firestore, SQLAlchemy, REST APIs, Celery

Frontend: React 18, Vite, TypeScript, Tailwind CSS, Next.js

DevOps and Tools: Docker, Docker Compose, Google Cloud Run, Git, GitHub, CI/CD, pytest, Locust, n8n, Power BI

PROFESSIONAL EXPERIENCE

AI Engineer

ShipCube (Remote) | Jan 2026 - Present

- Promoted to full-time AI Engineer after a three-month data science internship.
- Design and ship internal AI and automation services for 3PL logistics operations, spanning agentic LLM pipelines, RAG retrieval, cloud deployment and full-stack delivery.
- Built a production operations dashboard serving three role-based user groups, with JWT session management, role-based access control and role-specific landing views; React and TypeScript on the frontend, Node.js and Express on the backend.
- Engineered a natural-language assistant that interprets a question, routes it to the correct data source and returns a live answer or a generated report, with OAuth token handling that survives serverless cold starts.
- Replaced a low-code billing workflow with a deterministic Python service (FastAPI, Docker, Cloud Run) in which every charge is a pure function - deliberately keeping LLMs out of financial calculations so results are reproducible and auditable.
- Built an AI email bot and chatbot on an agentic pipeline using Hugging Face, LangChain and LLMs, with knowledge-base ingestion and RAG for context-aware understanding and dynamic reply generation.
- Own model fine-tuning, version control and CI/CD, improving pipeline reliability and maintainability.

Data Science Intern

ShipCube (Remote) | Oct 2025 - Dec 2025

- Built an AI email automation system using NLP and sentiment analysis to improve client communication efficiency.
- Engineered sentiment analysis and content categorisation models in Python, reaching 96% accuracy on held-out test data.

Design and Development Engineer (Intern)

Precision Electronics Limited | Jul 2024 - Sep 2024

- Automated data processing workflows in Python and Excel, cutting manual workload by 40% and improving data accuracy across reporting tasks.
- Built interactive Power BI dashboards for production and performance metrics, improving team decision-making efficiency by 35%.
- Produced technical documentation and analytical reports, increasing cross-team transparency by 25%.

PROJECTS

CreatorOS - Generative Video Platform

FastAPI, React, TypeScript, FLUX.1-schnell, Kokoro-82M, ffmpeg, PostgreSQL

- Built a 6,310-line Python API turning a single goal into a finished 1080x1920 vertical video across 8 generative stages: trends, research, script, frames, voice and music.
- Designed provider-agnostic LLM routing across Hugging Face, Ollama, vLLM and llama.cpp with automatic failover; every stage reports whether a model ran or a template substituted, so output provenance is never hidden.
- Cut generation cost by expanding only opened concepts, producing 200 screened concepts per keyword from one cheap pass instead of 200 full script generations.
- Ran Kokoro-82M text-to-speech locally at 0.8x realtime on CPU behind a 3-provider fallback chain, removing the API dependency and the per-render cost.

AI Research Agent - Agentic Research Platform

FastAPI, LangGraph, ChromaDB, Next.js, Celery, Redis, Docker

- Built an 8-stage agentic pipeline aggregating 13 verified research sources into fully cited reports exportable in 4 formats.
- Implemented hybrid retrieval combining vector search and BM25 with reciprocal rank fusion over ChromaDB, using sentence-aware chunking so citations never split mid-sentence.
- Added cross-source fact verification with conflict detection and per-claim confidence scoring, surfacing disagreement between sources rather than averaging it away.
- Abstracted 4 LLM providers behind automatic failover; the pytest suite mocks every external API so tests run with no network and no API keys.

High-Concurrency Flash Sale Ticket Booking System

FastAPI, PostgreSQL, Redis, asyncpg, Locust, Docker

- Engineered a booking API sustaining 500 concurrent users in load testing with 0 oversells and 0 database crashes.
- Eliminated double-booking using PostgreSQL row-level locking (SELECT ... FOR UPDATE), guaranteeing each of 100 tickets was sold exactly once under full contention.
- Absorbed roughly 99% of sold-out traffic in a Redis cache layer, rejecting requests in under 5 ms before they could reach PostgreSQL.

EmotionGesture AI - Real-Time Computer Vision Suite

OpenCV, MediaPipe, DeepFace, Streamlit, pytest

- Built a multi-modal pipeline recognising 8 hand gestures, counting 0-10 fingers across 2 hands and classifying 7 facial emotions from a single webcam feed.
- Removed frame flicker with a temporal stabiliser and a per-hand 1-Euro filter, and held frame rate by moving capture and emotion inference onto background threads.

Resume Screening App

scikit-learn, TF-IDF, KNN, NLTK, Streamlit

- Trained a K-Nearest Neighbours classifier on TF-IDF features reaching 98% test accuracy across 25 job categories.
- Shipped the model from notebook to browser with PDF and TXT parsing, text cleaning and inference behind a single Streamlit upload.

EDUCATION

Bachelor of Computer Applications (BCA) - IIMT University | 2025

Class XII, CBSE - Vidya Bal Bhawan Senior Secondary School - 78% | 2022